Total No. of Questions : 8]

SEAT No. :

PD4483

[Total No. of Pages : 2

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T.E. (Mechanical/Electrical) (Honors)

e-VEHICLE SYSTEM DESIGN

(2019 Pattern) (Semester - VI) (302033MJ)

Time : 2½ Hours] ***[Max. Marks : 70***

Instructions to the candidates:

1) ***Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Draw neat labeled diagrams wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Use of non-programmable electronic calculator is permitted.***
5) ***Assume Suitable/Standard data if necessary.***

Q1) a) Classify braking system used in automotive. **[8]**

b) Explain traction control system in EV. **[9]**

OR

Q2) a) Explain regenerative braking system. **[8]**

b) Explain traction control system in EV. **[9]**

Q3) a) Explain all wheel drive layout design. **[8]**

b) Explain Transmission system in e-Vehicle. **[9]**

OR

Q4) a) Explain Transmission system in e- Vehicle. **[8]**

b) What is open and locked differential system. **[9]**

Q5) a) Explain battery compartment and its need in E-Vehicle. **[9]**

b) Write short note on : **[9]**

i) Vent management system

ii) Pack cooling system

OR

Q6) a) Explain battery performance degradation modelling. **[9]**

b) Explain effect of Overcharging and Deep Discharging EV Batteries. **[9]**

Q7) a) Explain ergonomics base Roll cage with neat sketches. **[9]**

b) Explain the importance and process involved in Crash/Impact analysis. **[9]**

OR

Q8) a) Explain Vehicle dynamics and its components. **[9]**

b) Explain structural design aspect of roll cage/body frame. **[9]**

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